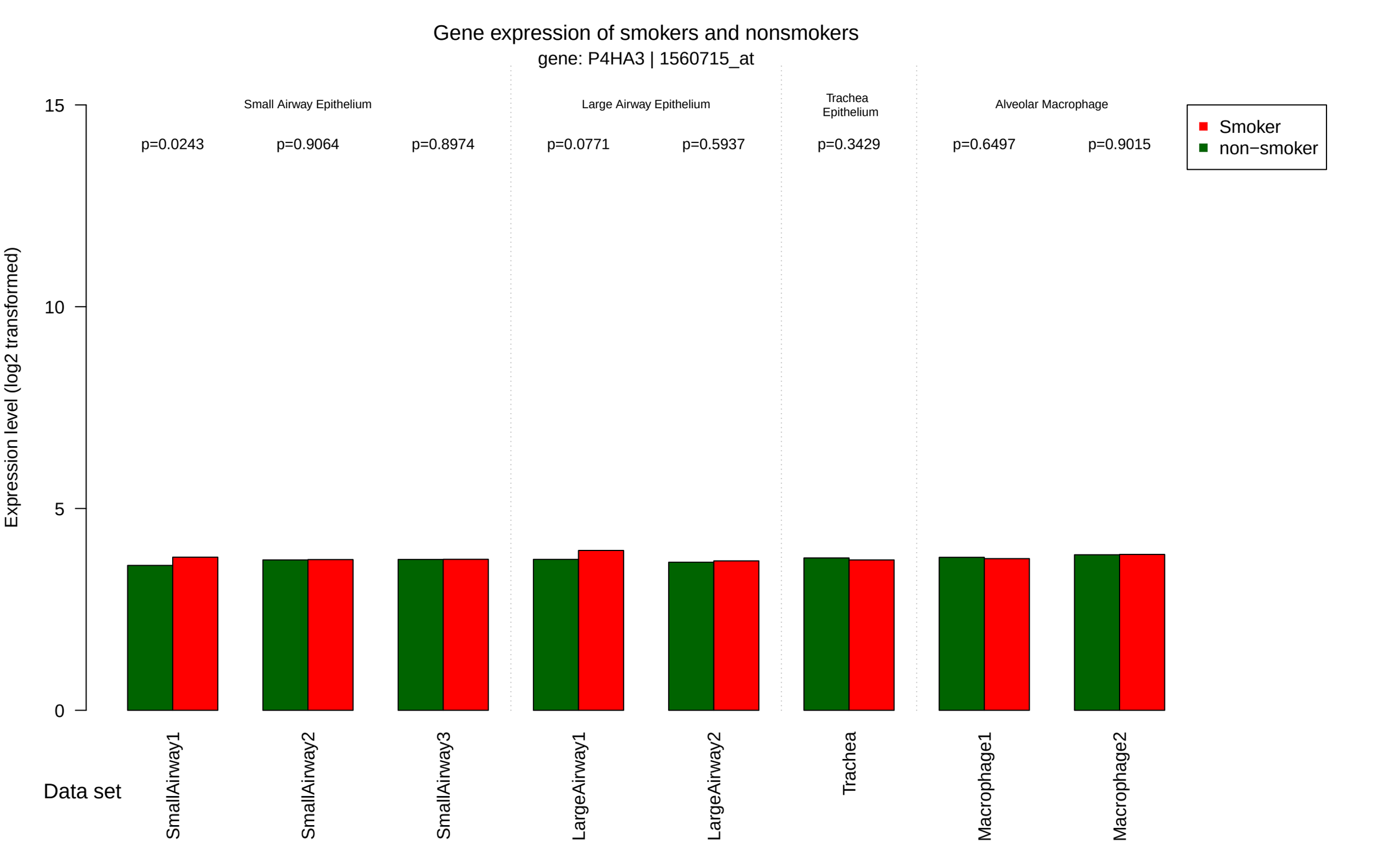
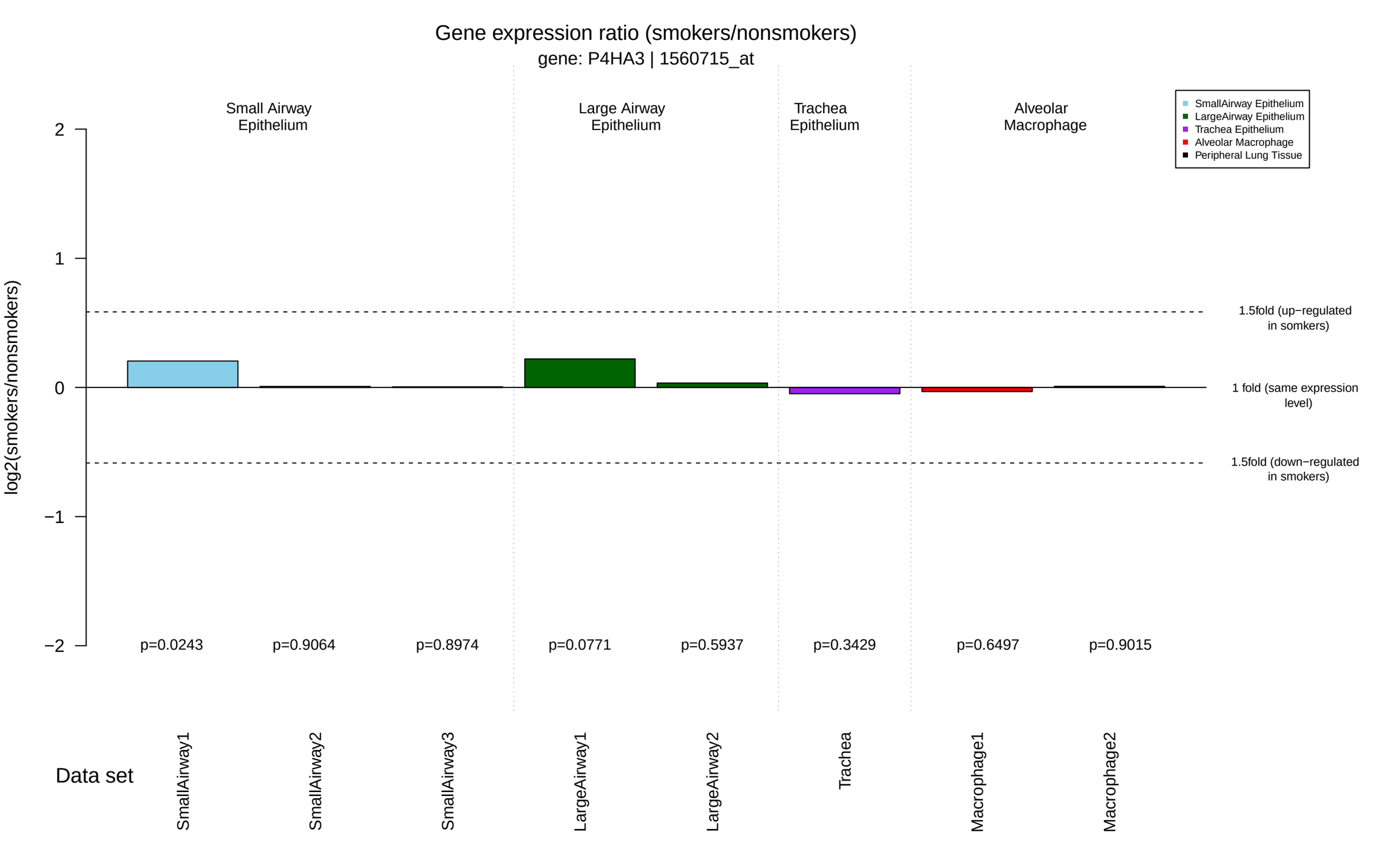
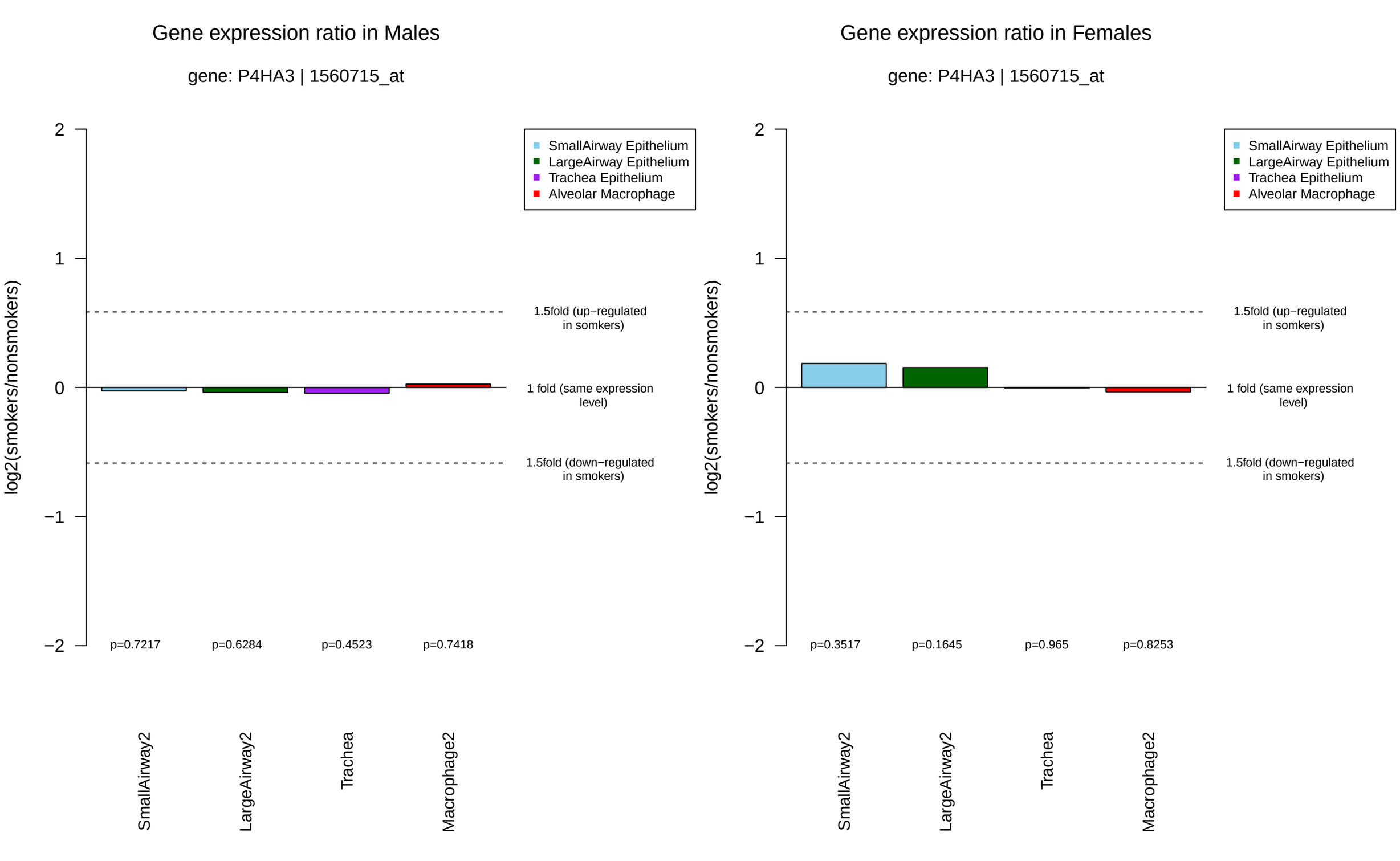
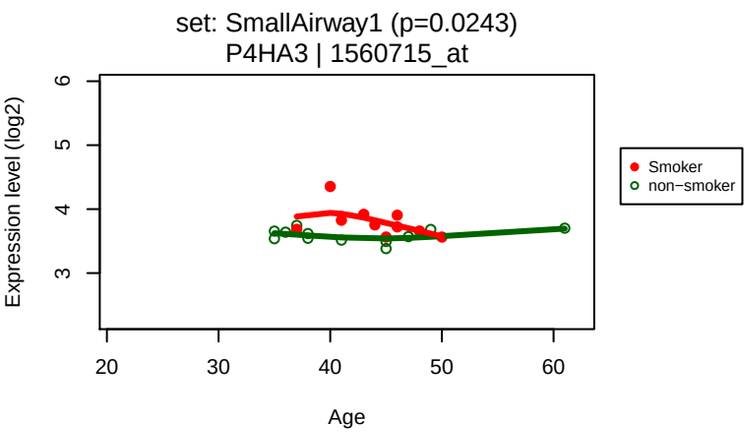
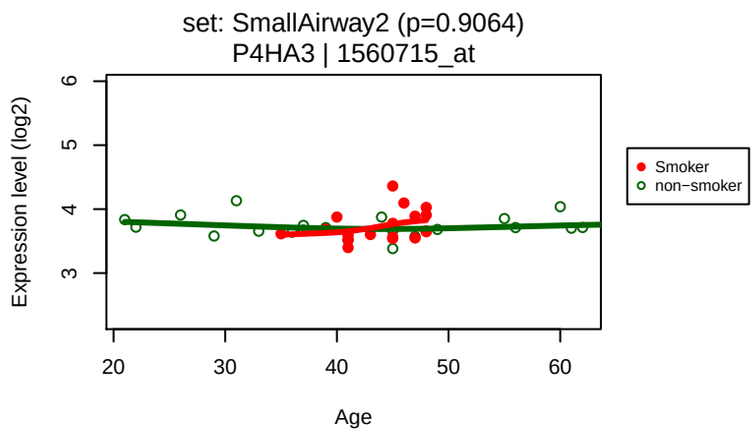
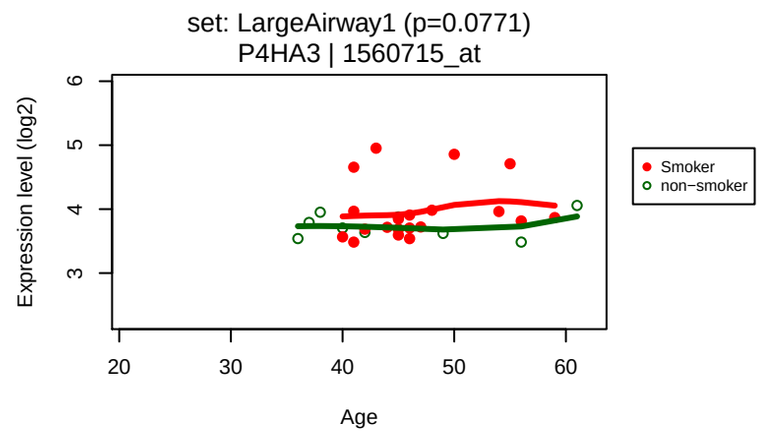

Gene expression of smokers and nonsmokers
gene: P4HA3 | 1560715_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0243
p=0.9064
p=0.8974
p=0.0771
p=0.5937
p=0.3429
p=0.6497
p=0.9015
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

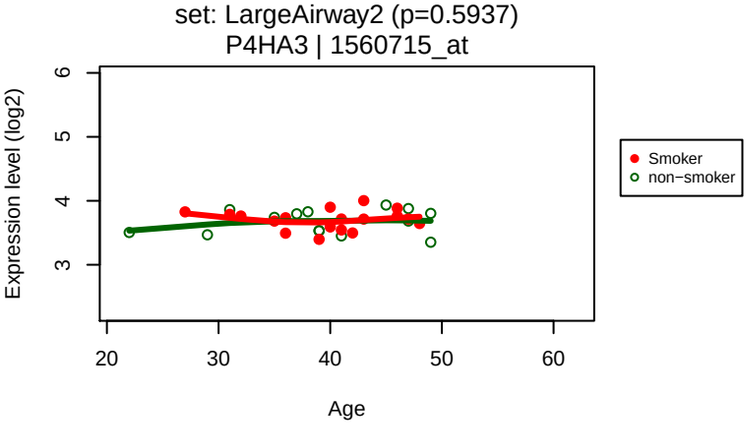

Gene expression ratio (smokers/nonsmokers)
gene: P4HA3 | 1560715_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0243
p=0.9064
p=0.8974
p=0.0771
p=0.5937
p=0.3429
p=0.6497
p=0.9015
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

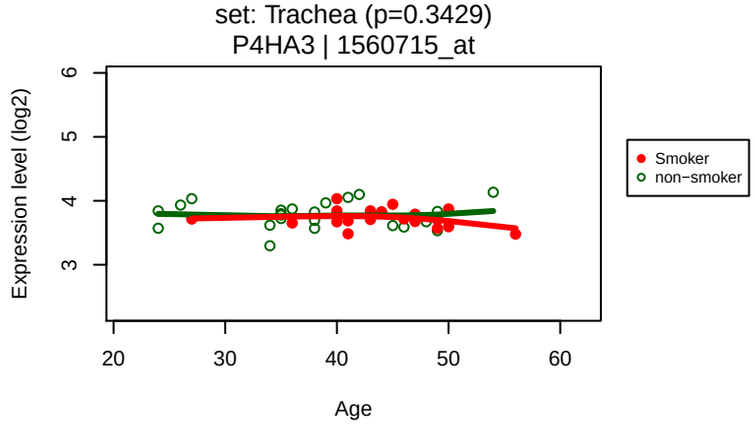

Gene expression ratio in Males
gene: P4HA3 | 1560715_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7217
p=0.6284
p=0.4523
p=0.7418
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: P4HA3 | 1560715_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3517
p=0.1645
p=0.965
p=0.8253
SmallAirway2
LargeAirway2
Trachea
Macrophage2

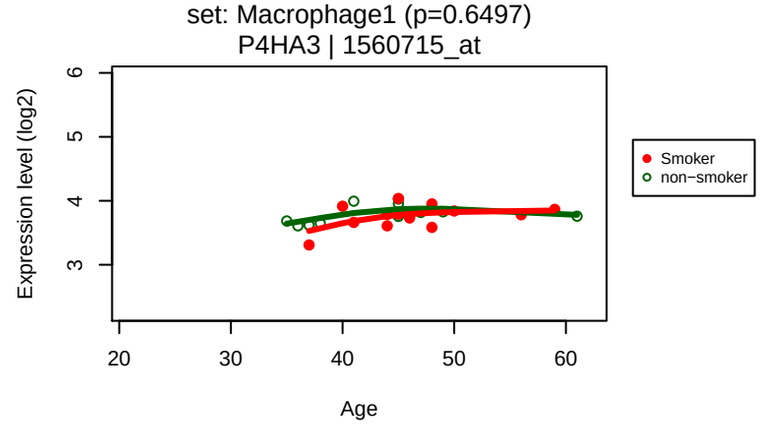

set: SmallAirway1 (p=0.0243)
P4HA3 | 1560715_at
Expression level (log2)
Age
Smoker
non−smoker

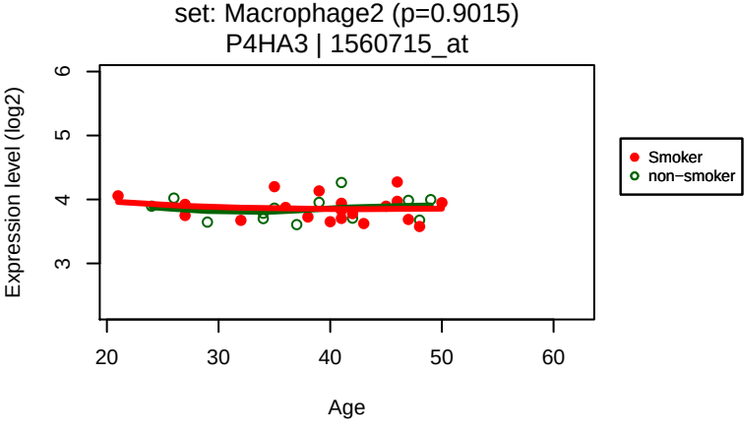

set: SmallAirway2 (p=0.9064)
P4HA3 | 1560715_at
Expression level (log2)
Age
Smoker
non−smoker

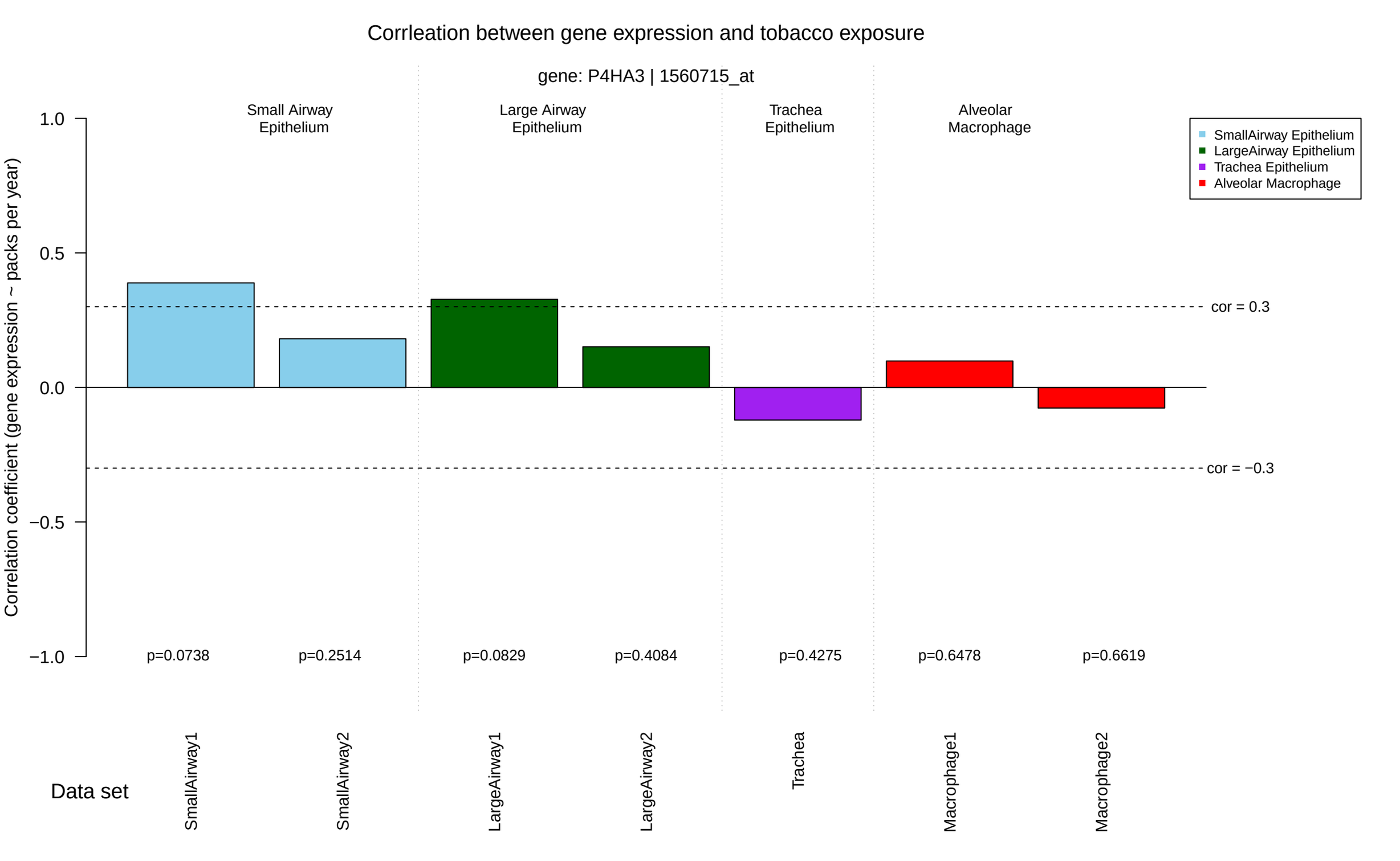

set: LargeAirway1 (p=0.0771)
P4HA3 | 1560715_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5937)
P4HA3 | 1560715_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3429)
P4HA3 | 1560715_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6497)
P4HA3 | 1560715_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9015)
P4HA3 | 1560715_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: P4HA3 | 1560715_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0738
p=0.2514
p=0.0829
p=0.4084
p=0.4275
p=0.6478
p=0.6619
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2